



Food safety in Thailand 2: Pesticide residues found in Chinese kale (*Brassica oleracea*), a commonly consumed vegetable in Asian countries



Sompon Wanwimolruk^{a,*}, Onnicha Kanchanamayoon^a, Kamonrat Phopin^{a,b}, Virapong Prachayasittikul^b

^a Center for Innovation Development and Technology Transfer, Faculty of Medical Technology, Mahidol University, Bangkok 10700, Thailand

^b Department of Clinical Microbiology and Applied Technology, Faculty of Medical Technology, Mahidol University, Bangkok 10700, Thailand

HIGHLIGHTS

- Significant pesticide residues were detected in Chinese kale sold in Thailand.
- MRL exceedance was found and this was higher than that seen in developed countries.
- Washing vegetables under running water can remove pesticide residues significantly.
- To be safe, washing vegetables before consumption is advisable.

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ABSTRACT

There is increasing public concern over human health risks associated with extensive use of pesticides in agriculture. Regulation of pesticide maximum residue limits (MRLs) in food commodities is established in many developed countries. For Thailand, this regulation exists in law but is not fully enforced. Therefore, pesticide residues in vegetables and fruits have not been well monitored. This study investigated the pesticide residues in Chinese kale, a commonly eaten vegetable among Asians. The Chinese kale samples ($N = 117$) were purchased from markets in Nakhon Pathom Province, Thailand, and analyzed for the content of 28 pesticides. Analysis was performed by the multiresidual extraction followed by GC–MS/MS. Of pesticides investigated, 12 pesticides were detected in 85% of the Chinese kale samples. Although carbaryl, deltamethrin, diazinon, fenvalerate and malathion were found in some samples, their levels were lower than their MRLs. However, in 34 samples tested, either carbofuran, chlorpyrifos, chlorothalonil, cypermethrin, dimethoate, metalaxyl or profenofos was detected exceeding their MRLs. This represents a 29% rate of pesticide detection above the MRL; a rate much higher than in developed countries. Washing vegetables under running water significantly reduced ($p < 0.05$) profenofos residues by 55%. The running water method did not significantly decrease cypermethrin residues in the samples but washing with vinegar did. Our research suggests that routine monitoring of pesticide residues is necessary to reduce the public health risks associated with eating contaminated vegetables. Washing vegetables before consumption is advisable as this helps to reduce the level of pesticide residues in our daily intake.

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1. Introduction

Pesticides are chemicals that help protect plants against molds, fungi, rodents, and insects. They help prevent crop loss and, potentially, human disease. Numerous pesticides are widely used all over the world. According to the Environmental Protection Agency (EPA), there are currently more than 865 registered pesticides. It is not uncommon to detect pesticide residues in vegetables and fruits. In recent years we have seen a substantial increase in the importance placed on aspects of pesticide residues and growing tension for better agricultural practices (Payá et al., 2007;

Yang et al., 2014). Accordingly many control authorities in various countries have established maximum residue limits (MRLs) or tolerances to protect consumer's health and also the environment (USDA, 2014; European Commission, 2015). The MRL is a product limit, not a safety limit. The MRL is the maximum concentration of a pesticide residue (expressed as mg/kg) recommended by the Codex Alimentarius Commission to be legally permitted in or on food commodities and animal feeds (Codex Alimentarius Commission, 2015; European Commission, 2008, 2015). MRLs are based on supervised residue trial data where the pesticide has been applied in accordance with GAP (Good Agricultural Practice). By implementing dietary intake calculations and comparing the intake to toxicological reference values it is possible to verify whether foods derived from commodities that comply with the respective MRLs are safe for consumers. If this is not the case, the MRL does not go into

* Corresponding author.

E-mail addresses: sompon-999@hotmail.com, sompon.wan@mahidol.ac.th (S. Wanwimolruk).

the legislation. Therefore, foods containing residues below or at the level of the MRL are safe for consumption. However, foods containing residues above the level of the MRL are not inevitably unsafe. In many cases there is rather a margin between the calculated dietary intake and the toxicological reference value. Consideration of the various dietary residue estimates and determinations both at the national and international levels in comparison with the acceptable daily intake (ADI), should indicate that foods complying with Codex MRLs are safe for human consumption (Codex Alimentarius Commission, 2015).

For developing countries such as Thailand, some regulations on MRL in food commodities exist, but are often not fully enforced or promulgated. Pesticides have been widely used in agriculture in Thailand (Harnpicharnchai et al., 2013). The most common categories of pesticides imported into Thailand are herbicides, followed by insecticides and fungicides (Sapbamrer and Nata, 2014). In the midst of the insecticides, organophosphates and carbamates are highly popular for guarding crops from insects' invasion. The usage of pesticides in agriculture has been related with occupational health of farmers, gardeners and consumers.

Chinese kale (*Brassica oleracea*) is also known as Chinese broccoli. It is a leaf vegetable featuring thick, flat, glossy blue-green leaves with thick stems and a small number of tiny, almost vestigial flower heads similar to those of broccoli. Flavor of Chinese kale is very similar to that of broccoli, but somewhat bitterer. The flavor of Chinese kale is also markedly stronger than broccoli. Chinese kale is used widely in Chinese cuisine, and especially in Cantonese cuisine. In Thailand, a number of popular Thai dishes have Chinese kale as a main ingredient; and in some dishes, Chinese kale is consumed fresh, without cooking. This has potential for toxicity if the vegetable is consumed freshly and daily. In fact, data on pesticide contamination in vegetables in Thailand is scarce and systemic investigation is needed to verify the current status of pesticide contamination in foods, especially in vegetables and fruits. The present study took an initial effort to determine pesticide residues in a commonly consumed vegetable, Chinese kale. Moreover, appropriate and effective washing methods to remove pesticides from Chinese kale were evaluated.

2. Materials and methods

2.1. Chemicals and standards

All organic solvents used were HPLC grade. Anhydrous magnesium sulfate, sodium chloride, primary and secondary amines (PSA, particle size 40 μm), graphited carbon black (GCB) and C18 sorbent (particle size 40 μm) were obtained from Supelco (Sigma-Aldrich Corp., St. Louis, USA). Acetonitrile was purchased from Merck (Darmstadt, Germany). Twenty eight pesticides and two metabolite standards including aldrin, atrazine, captan, carbaryl, carbofuran (and its two metabolites carbofuran-3-hydroxy and carbofuran-3-keto), carbosulfan, chlormefos, chlorpyrifos, chlorothalonil, cypermethrin, deltamethrin, diazinon, dichlorvos, dicofol, dimethoate, ethion, fenitrothion, fenvalerate, lambda-cyhalothrin, malathion, metalaxyl, methidathion, methomyl; paraoxon-methyl, phosalone, pirimicarb, pirimiphos-methyl and profenofos were purchased from Dr. Ehrenstorfer (Augsburg, Germany). These pesticides studied were selected on the basis of their wide use in agriculture in Thailand. Purity of these pesticide standards was >98%. Individual stock of standard solutions (1000 mg/L) was prepared in acetonitrile. Working surrogate spiking standard solutions were prepared by an appropriate dilution of the stock solutions with acetonitrile. These standard solutions were protected from light and kept at -20°C until required.

2.2. Chinese kale samples

Chinese kale samples ($N = 117$) were purchased randomly from 12 local markets located in Nakhon Pathom Province near Bangkok,

Thailand. The study was conducted during a period from October 2013 to July 2014. These Chinese kale samples were not 'organic produce'. Approximately 500 g of vegetables were purchased and the samples were brought to the laboratory for analysis which was performed within 24 h. For washing experiments (refer to the next section), Chinese kale samples (approximately 6 kg each) were bought and screened for presence of pesticides. Those samples with high concentrations of pesticides detected were further subjected to washing experiments. The samples for washing experiments were divided into five portions: one portion was processed as such (unwashed); second portion after washing with running water, third portion after washing with NaHCO_3 , fourth portion after washing vinegar, and fifth portion after washing with KMnO_4 . There were 8 samples in each washing experiment group ($N = 8$). The representative portion (150–200 g) of the vegetable sample was cut into small pieces and blended using a food processor and mixed thoroughly. The homogenized samples were then extracted and processed as described in following section.

2.3. Washing experiments

Chinese kale samples which were detected for high concentration of two pesticides namely profenofos and cypermethrin, were subjected to washing experiments. This was specifically to compare effectiveness of four different washing methods. These included running water method, washing by using NaHCO_3 , vinegar and KMnO_4 . These four washing methods are widely used for washing vegetables and fruits by Thai people (Hatthakosol, 2013; Thai Pharmacist On Web). It is common for Thai families to have NaHCO_3 , vinegar and KMnO_4 on hand for washing their vegetables and fruits. Running water method was carried out by washing the vegetables (approximately 200 g) under running tap water for 2 min with gentle rubbing with hands while washing and the water was discarded. Washing with NaHCO_3 , the vegetables were soaked in 5 L of diluted NaHCO_3 solution (1/2 teaspoon of NaHCO_3 in 5 L of water) for 10 min. Then, the solution was discarded and the vegetables were further rinsed with 5 L of water. Washing with vinegar, the vegetables were soaked in 5 L of diluted vinegar solution (1/2 teaspoon of cooking vinegar in 5 L of water) for 10 min. The solution was discarded and the vegetables were rinsed with 5 L of water. For washing with KMnO_4 , the Chinese kale samples were soaked in 5 L of diluted KMnO_4 solution (100 mg KMnO_4 in 5 L of water) for 10 min. Then, the solution was discarded and the vegetables were washed with 5 L of water. All washed samples were kept on blotting paper to remove the excess water. Subsequently, the vegetable samples were extracted, cleaned up in a similar manner as the fresh Chinese kale samples, were processed and analyzed.

2.4. Analytical methods

The analysis of pesticide residues was carried out using the pesticide multiresidue QuEChERS (Quick Easy Cheap Effective Rugged and Safe) method as described previously (Anastassiades et al., 2003; Lehotay, 2007; Lehotay et al., 2010; Payá et al., 2007). Briefly, extraction of pesticides was performed by extracting 15 g of homogenized vegetable (Chinese kale) with 15 mL acetonitrile saturated with 6 g of magnesium sulfate and 1.5 g of sodium chloride. This extraction process was followed by a cleaning up procedure. This was performed by transferring the supernatant into another tube containing 50 mg of primary–secondary amines (PSA), 7.5 mg graphited carbon black (GCB) and 150 mg magnesium sulfate. After agitation and centrifugation, the extract supernatant was then transferred to an autosampler vial for direct injection into the Bruker GC-MS/MS system. Recovery studies for method validation were performed by adding appropriate volumes of working solution to blank samples. The method validation with respect to recovery, reproducibility, calibration linear range, limit of detection (LOD), and limit of quantification (LOQ) was carried out for Chinese kale matrix as described previously (Koesukwiwat et al., 2011; Dong et al., 2012).

Detection of pesticides was performed using a Bruker 456 gas chromatography (GC) coupled with Bruker Scion Triple Quadrupole mass spectrometer (GC–MS/MS). Details of GC–MS/MS conditions were referred to as in the previous report (Duff and Voglino, 2012). Multiple reaction monitoring (MRM) acquisition method and two ion transitions at the experimentally optimized collision energy (CE) were monitored for each pesticide analyte.

2.5. Calibration and quantification

Calibration curves of each pesticide of interest were performed according to the established procedure (Koesukwiwat et al., 2011; Lehotay, 2007; Lehotay et al., 2010). These were conducted using an identical procedure each time when a new unknown sample set was analyzed. The ions selected for quantification in the GC–MS/MS analysis had the mass-to-charge ratio (m/z) of the most abundant ions derived from the ionization process of each pesticide, while the remaining ions were used for confirmation of the analyte identity. Aldrin was used as an internal standard. The ratio of the peak area of the pesticide standard to that of the internal standard was employed for

quantification. Excellent linearity and reproducibility of calibration curves were obtained as illustrated by the coefficient of determination (r^2) values of >0.93 and relative standard deviations (RSD) of $<20\%$. The signal-to-noise ratio of pesticides of interest measured at 0.01 ppb was well above 30 for all pesticides studied. Therefore, detection limits were below 0.01 ppb using the sample preparation procedures described above. Determination of pesticides in an unknown vegetable sample was done in duplicate unless otherwise stated. Blank samples of Chinese kale used for recovery experiments and for preparation of the matrix-matched multi-level calibration standards were samples previously assayed and shown to be free of the pesticides of interest. MRL values for each pesticide in the Chinese kale were cited from recommended MRL values established by the Thailand National Bureau of Agricultural Commodity and Food Standards (2008) and Codex Alimentarius Commission (2009).

2.6. Statistical analysis

All data are presented as mean $\pm$ standard deviation (SD). Differences between groups were assessed by a one-way analysis of variance

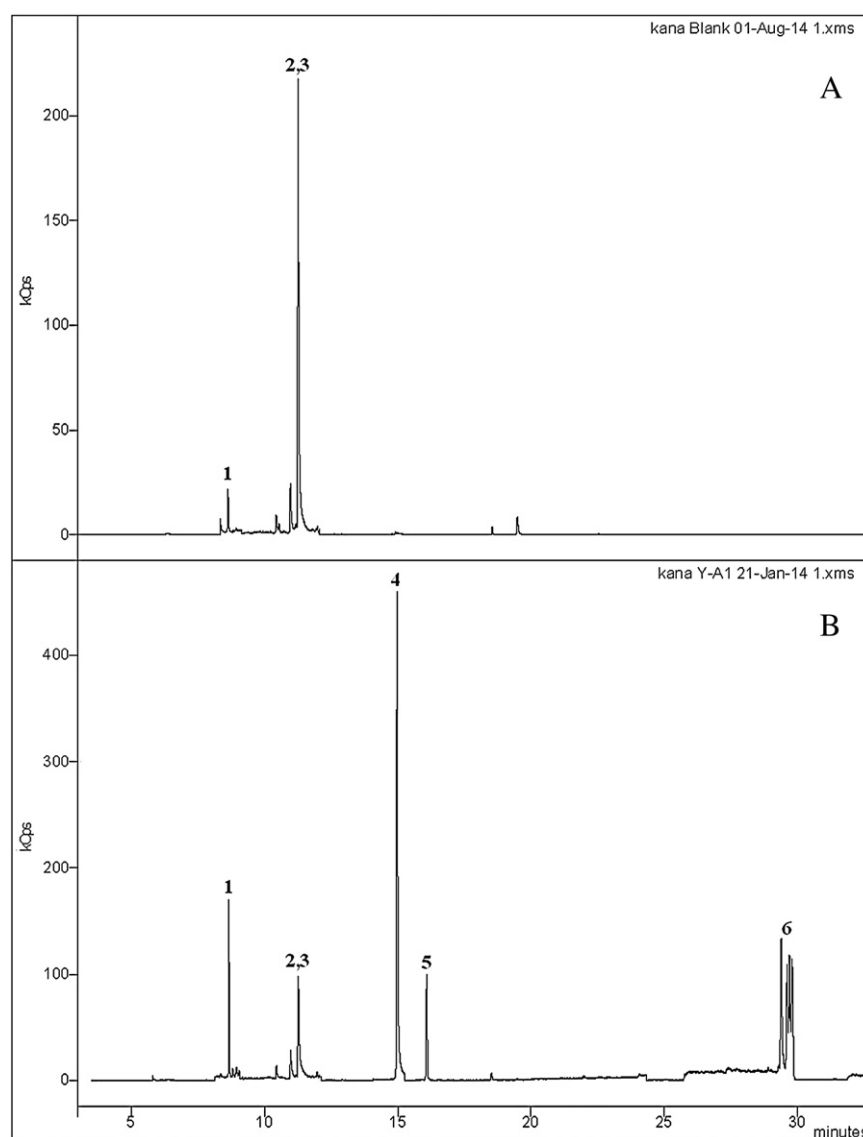


Fig. 1. The chromatograms of blank extraction of Chinese kale sample (A); and a representative chromatogram of a Chinese kale sample extract containing two pesticides, metalaxyl and cypermethrin (B). Peak identification: (1) unknown peak from plastic centrifuge tube; (2) and (3) endogenous compounds found in Chinese kale samples; (4) metalaxyl; (5) aldrin (internal standard); and (6) cypermethrin.

(ANOVA), followed by multiple comparisons utilizing Tukey's test. Probability values of less than 0.05 were considered as statistically significant. SPSS version 18.0 (SPSS Inc., Chicago, IL, USA) was used for statistical analysis.

3. Results

3.1. Verification of the analytical method used

Twenty eight pesticides and two metabolites of carbofuran were studied. These pesticides were selected from among those widely used in the agriculture of vegetables and fruits in Thailand. Under conditions employed for GC–MS/MS method, all pesticides studied were satisfactorily separated with high sensitivity and selectivity. The absence of co-extracted interferences for all varieties of Chinese kale was verified by blank extract analysis (Fig. 1A). The GC–MS chromatogram of blank Chinese kale extract showed that there were two significant endogenous compounds (peaks 2 and 3) eluted at retention times (RT) of 10.99 and 11.27 min (Fig. 1A). These two endogenous peaks were observed in every sample of Chinese kale extract. However, the mass and mass ratios of these peaks were not the same as those of the pesticides of interest. These two peaks were unable to identify the chemical structures based on the current GC–MS/MS program for the NIST/EPA/NIH Mass Spectral Library. This suggests that the observed two peaks in the extracts of Chinese kale are endogenous compounds that occurred naturally in Chinese kale. Of note, in the GC–MS/MS chromatograms of Chinese kale extracts, there was an abundant peak, peak 1 eluting with RT of 8.64 min (Fig. 1A and B). This peak is almost co-eluted with one of the pesticide studied, i.e., carbaryl which was chromatographically eluted with RT of 8.82 min. QuEChERS extraction experiments were carried out using polypropylene plastic centrifuge tubes (50 mL, Jet Bio-Filtration Products Co. Ltd., Guangzhou, China) under the same conditions. It revealed that this interfering peak (peak 1, RT of 8.64 min) was derived from the plastic centrifuge tube used. It was extractable by acetonitrile that was used for sample preparation. We chose to use these polypropylene plastic centrifuge tubes due to economic considerations. These plastic centrifuge tubes are much cheaper (US\$0.26 vs US\$47 per tube) than the Teflon centrifuge tubes recommended in the original QuEChERS extraction method (Anastassiades et al., 2003; Lehotay, 2007). This obstruction because of unknown interfering peak from the plastic tubes had no impact on quantification of carbaryl as the interfering peak had different mass and mass ratios in MS analysis. Moreover, in all Chinese kale samples tested, there were no identifiable peaks detected with the same RT as aldrin which was used as an internal standard in our GC–MS/MS assay. This verifies the validity of using aldrin as the internal standard for the current assay. Representative chromatogram of Chinese kale sample after extraction is illustrated in Fig. 1B. This particular sample had typical endogenous peaks (peaks 2 and 3) and contained two pesticides, metalaxyl (peak 4) and cypermethrin (peak 6). Of note, cypermethrin was eluted as four multiple peaks under this GC–MS/MS condition. Peak 5 is an internal standard (aldrin) used for the GC–MS/MS assay.

The limits of quantification (LOQs) were established for each pesticide of interest. The LOQ was 0.5 ppb (0.5 µg/kg) for aldrin, atrazine, carbaryl, carbofuran (and its two metabolites carbofuran-3-hydroxy and carbofuran-3-keto), chlorothalonil, chlorpyrifos, cypermethrin, deltamethrin, diazinon, dichlorvos, dicofol, dimethoate, malathion, metalaxyl, methidathion, phosalone, pirimicarb and profenofos; and 10 ppb (10 µg/kg) for captan, carbosulfan, chlormefos, ethion, fenitrothion, fenvalerate, lambda-cyhalothrin, paraoxon-methyl and pirimiphos-methyl. At these LOQs, the recovery and precision of the assay were satisfactory, i.e., 72–118% for recovery and <20% RSD for precision. Reproducibility of the assay was evaluated by analyzing the concentration of pesticides studied in the Chinese kale samples at two different concentrations (low and high). The results have demonstrated that the reproducibility of the analytical assay was good. For example,

with profenofos as an analyte, the reproducibility ($N = 5$) at low concentration (0.5 ppb) was 11% and at a high concentration (5000 ppb) was 6%. The overall reproducibility data for the 28 pesticides studied ranged from 4 to 16%. These are in line with the criteria established for analysis of pesticide residues in foods (European Commission, 2009). Thus, these indicate that the current GC–MS/MS used is practicable to analyze the pesticides of interest.

3.2. Pesticide tested versus pesticide residues detected, and differences between markets

The present study was conducted in Nakhon Pathom Province which is a central province near Bangkok, Thailand. This province is known for agriculture produces including many fruits and vegetables. Fresh samples of Chinese kale were randomly purchased from 12 local markets in four districts of Nakhon Pathom Province, namely Phuthamonthon (markets A–F), Samphan (markets G, H, I), Muang Nakhon Pathom (markets J and K) and Nakhonchaisri (market L). Results obtained, as shown in Fig. 2A illustrate, the frequency of detected pesticides in Chinese kale samples tested for each market. In individual markets, the proportion of samples was categorized according to the relative contents of pesticides found in the samples. Pesticides were detected in samples of Chinese kale from all the markets investigated. The incidence of detected pesticide >MRL was highest in markets G, H and I (50%), although sample size in these markets was small ($n = 4$). Among those markets with reasonable large sample size ($n = 13–20$), the incidence of having pesticide detected of >MRL was noticeably high in market A (50%), market D (27%), and market B (23%). Fig. 2B summarizes the proportions of Chinese kale samples categorized by incidence of detected pesticides, i.e., no pesticides; pesticide detected of <MRL; and pesticide detected of >MRL. As it can be seen in this figure, the overall detected pesticide rate was 85% (100/117). The Chinese kale samples with no pesticides detected were 15%. Among the pesticide contaminant samples, 34 samples (29%) had pesticide residues equal to or exceeding the MRL (>MRL, 34/117), whereas 56% of the contaminant samples contained pesticide residues of less than the MRL (<MRL, 66/117).

3.3. Pesticide detection rates and MRL exceedances

Of the 28 pesticides studied, 12 different individual pesticides (i.e., not including metabolite of carbofuran) were detected in the fresh Chinese kale samples collected from local markets in Nakhon Pathom Province (Fig. 3). These included carbaryl, carbofuran, chlorpyrifos, chlorothalonil, cypermethrin, deltamethrin, diazinon, dimethoate, fenvalerate, malathion, metalaxyl and profenofos. Some samples contained only one pesticide but some 65% (76 out of 117) of the samples had multiple residues of more than one pesticide found in the same sample. Table 1 summarizes the MRL values used and the concentration ranges of pesticides detected in this study. Carbaryl was detected in 11% (13/117) of the Chinese kale samples tested (Fig. 3). Levels of carbaryl found in these samples ranged from 1 to 10 ppb which did not exceed the MRL value for carbaryl (50 ppb). Carbofuran and its metabolite carbofuran-3-keto were found in seven Chinese kale samples (Fig. 3). The occurrence of carbofuran was 6% (7/117). Levels of carbofuran detected in these samples ranged from 10 to 62,500 ppb. Two samples had carbofuran levels greater than its MRL (20 ppb). Four of the Chinese kale samples tested were found to contain chlorpyrifos. The levels of chlorpyrifos found ranged from 7 to 37,700 ppb; two of these samples exceeded the recommended MRL value (50 ppb).

Three of the samples had considerable levels of chlorothalonil (Fig. 3). The levels were ranged from 250 to 16,200 ppb. Of these three samples, one Chinese kale sample had a remarkable high level of chlorothalonil of 16,200 ppb which well exceeded its MRL (4000 ppb). Of interest, cypermethrin was the detected pesticide in the fresh Chinese kale samples and many of these samples exceeded the MRL value. As shown in Fig. 3, fifteen Chinese kale samples were found to

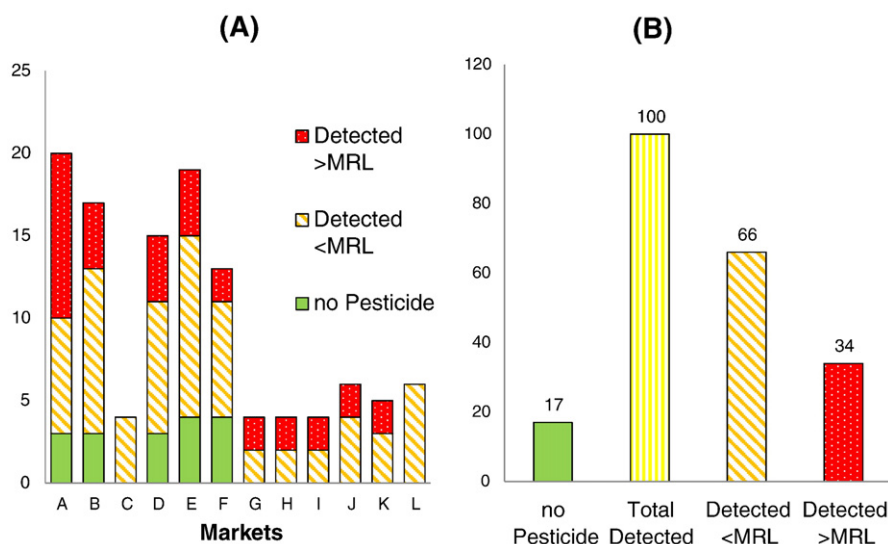


Fig. 2. Frequency distribution of (A) Chinese kale samples ($N = 117$) with no pesticide detected, detected pesticide residues with levels of $<MRL$, and with levels of $>MRL$ in 12 local markets investigated; and (B) Overall incidence of detected pesticide residues in Chinese kale samples: with no pesticide detected, total number of samples with detected pesticide residues, detected pesticide residues with levels of $<MRL$, and detected pesticide residues with levels of $>MRL$.

contain cypermethrin ranging from 84 to 83,100 ppb, in which four samples (27%, 4/15) contained cypermethrin levels of $>MRL$ value (1000 ppb). Deltamethrin was found in 5 samples of all 117 Chinese kale samples tested representing only 4% occurrence rate. Levels of deltamethrin detected in these samples ranged from 40 to 86 ppb which were lower than its MRL value (500 ppb). Diazinon is one of the pesticides studied that is found quite often in the Chinese kale samples tested. Its rate of occurrence in Chinese kale samples was 42% (49/117). However, concentrations of diazinon found in the samples ranging from 0.5 to 30 ppb, were lower than its MRL (50 ppb).

Metalaxyl and dimethoate were the most frequently found pesticides in the Chinese kale samples tested. Occurrence rate of these pesticides in the Chinese kale was 74% (87/117) and 41% (48/117) for metalaxyl and dimethoate, respectively. Levels of dimethoate detected in the Chinese kale samples were in the range of 0.5 to 239 ppb. Among these samples, 16 of them contained dimethoate exceeding

the MRL value (20 ppb). This represents a rate of greater than dimethoate's MRL of 33%. Metalaxyl detected in the Chinese kale samples investigated was in a range from 0.5 to 3920 ppb. Two of these 80 samples detected with metalaxyl exceeded the MRL value (2000 ppb).

Three other pesticides were also found in the Chinese kale samples. These were fenvalerate, malathion and profenofos. Fenvalerate and malathion were detected in the Chinese kale with low occurrence rate of 4% (5/117) and 7% (8/117), respectively. Notably, profenofos was found in the Chinese kale samples at a relative high detection rate of 25% (29/117). These samples contained profenofos ranging from 0.5 to 183,000 ppb, in which 13 of these samples had profenofos level higher than the recommended MRL (50 ppb). Some other 16 pesticides studied were not detected in the Chinese kale samples tested. These include aldrin, atrazine, captan, carbosulfan, chlormefos, dichlorvos, dicofol, ethion, fenitrothion, lambda-cyhalothrin, methidathion, methomyl, paraoxon-methyl, phosalone, pirimicarb and pirimiphos-methyl. The Supplemental information provides further details of pesticides detected in each sample of Chinese kale studied.

3.4. Effect of washing

Effect of washing on the removal of pesticide residues from the Chinese kale was investigated. Four different popular washing methods were compared for their effectiveness in removing two pesticides that

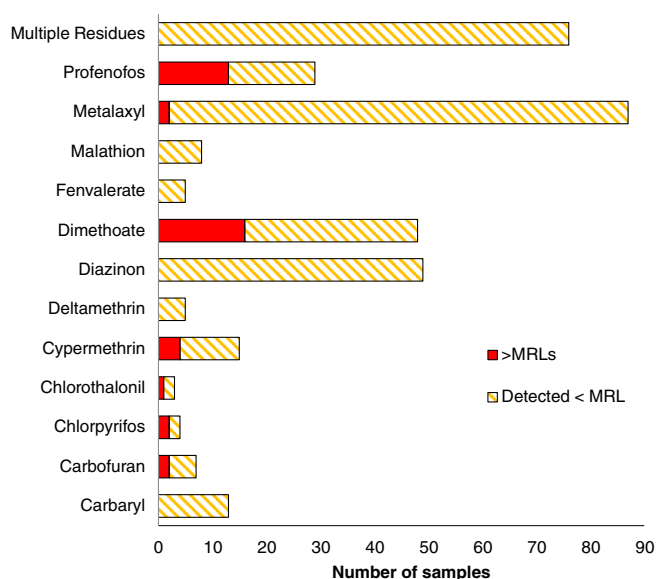


Fig. 3. Types of pesticides detected in the Chinese kale samples ($N = 117$). For each pesticide found, the samples were categorized according to the level of pesticide detected, i.e., pesticide detected with level of greater than MRL (■) and pesticide detected with level of less than MRL (▨).

Table 1
The MRL values used and pesticide concentration ranges found in this study.

Pesticide	MRL value (ppb)*	Concentration range (ppb) observed in Chinese kale samples
Carbaryl	50	1–10
Carbofuran	20	10–62,500
Chlorpyrifos	50	7.2–37,700
Chlorothalonil	4000	250–16,200
Cypermethrin	1000	84–83,100
Deltamethrin	500	40–86
Diazinon	50	0.5–30
Dimethoate	20	0.5–239
Fenvalerate	10,000	320–779
Malathion	3000	2–330
Metalaxyl	2000	0.5–3,920
Profenofos	50	0.5–183,000

* Sources: Codex Alimentarius Commission (2009); Thailand National Bureau of Agricultural Commodity and Food Standards (2008).

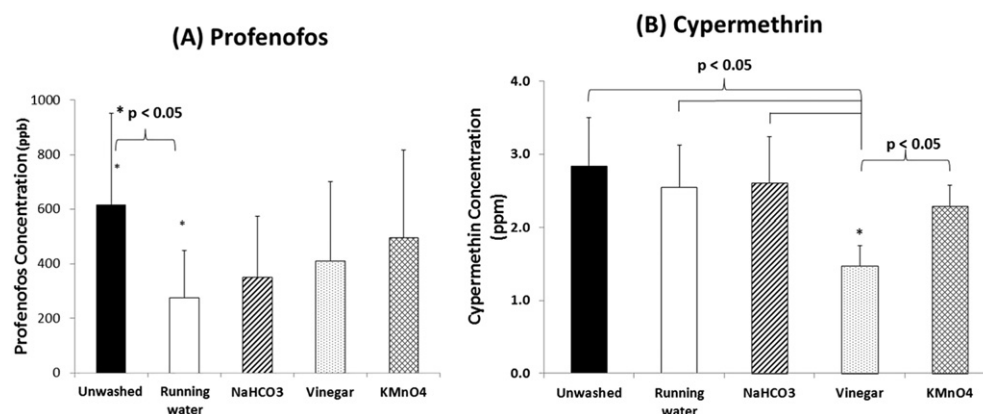


Fig. 4. Effects of different washing methods on removal of two pesticide residues from the Chinese kale samples: (A) profenofos and (B) cypermethrin. Details of each individual washing method are referred to [Materials and methods](#) section. The values given are means and standard deviations (N = 8).

were found in the Chinese kale samples tested. Fig. 4A shows changes in profenofos concentration after washing Chinese kale samples with different methods. All the four investigated washing methods caused a decrease in profenofos concentration after completion of washing processes. However, the differences produced by washing with NaHCO₃, vinegar and KMnO₄ were not significant as compared to the control unwashed samples (Fig. 4A). Only washing under running water significantly reduced ($p < 0.05$) the concentration of profenofos when compared with the unwashed samples. Mean concentration of profenofos after washing with running water method was 45% of the control unwashed vegetables. Experiments with other pesticide cypermethrin revealed different results (Fig. 4B). Washing Chinese kale samples with running water, NaHCO₃ and KMnO₄ methods did not significantly reduce the concentration of cypermethrin in the Chinese kale samples. In contrast, washing the vegetable samples with vinegar caused a significant decrease ($p < 0.05$) in cypermethrin concentration in the Chinese kale samples as compared with those control unwashed samples (Fig. 4B). The mean cypermethrin concentration in the vinegar washed samples was approximately 52% of those control unwashed samples. This was also significantly ($p < 0.05$) less than those washed with running water, NaHCO₃ and KMnO₄ methods.

4. Discussion

4.1. GC–MS analytical method validation

Development of GC–MS/MS analytical procedures, including sample preparation and GC–MS/MS analysis, was based on previous reports (Anastassiades et al., 2003; Koesukwiwat et al., 2011; Lehotay, 2007; Lehotay et al., 2010). The sample preparation involved solvent extraction and cleaning up process was carried out in accordance with the popular sample preparation, QuEChERS procedure for determination of pesticides (Anastassiades et al., 2003; Lehotay, 2007; Lehotay et al., 2010). The detection and quantitation of pesticides were successfully executed by the uses of GC coupled with quadrupole MS technique. This provided selective and sensitive assay to quantify individual pesticide in a mixture matrix such as vegetables and fruits. The current developed GC–MS/MS method was validated and appeared to provide good reproducibility, sensitivity, linear calibration curves and accuracy. This assay was proven to be suitable and applicable to determine the content of many pesticides in a leafy vegetable such as Chinese kale.

4.2. Pesticide detected in Chinese kale samples and incidence of pesticide detection

In this study, various types of pesticides were found to contaminate the Chinese kale, one of widely consumed vegetables in Asian countries.

These included carbamates (carbaryl, carbofuran), organochlorines (chlorothalonil), fenvalerate, metalaxyl, organophosphorus pesticides (chlorpyrifos, diazinon, dimethoate, malathion, profenofos), and pyrethroids (cypermethrin, deltamethrin). This implies that pesticides are used widely and extensively in the cultivation of Chinese kale in Thailand. This finding is consistent with those previously observed pesticide contamination in vegetables both in Thailand and other Asian countries (Chang et al., 2005; Sapbamrer and Hongsibsong, 2014; Swarnam and Velmurugan, 2013). A recent study conducted in northern part of Thailand (Sapbamrer and Hongsibsong, 2014) found that vegetables collected from markets containing organophosphorus pesticides greater than MRLs. These included garlic, Chinese cabbage, Vietnamese coriander, spring onion and Chinese kale. The finding with regard to Chinese kale coincides with our observation in which the levels of three organophosphorus pesticides (chlorpyrifos, dimethoate and profenofos) were greater than the recommended MRL.

Even though the pesticides were detected in Chinese kale samples from all the local markets investigated, the incidence of detected pesticide >MRL seemed to vary among the markets (Fig. 2A). Definite claims could not be made as the sample sizes in some markets were rather small. Variation in the incidence of detected pesticide >MRL may relate to the sources (or farms) where the vegetables were grown, differences in pesticides usage, and lack of GAP knowledge. Traceability of the produces was difficult to achieve and ultimately this was not the primary goal of the present study. We did ask the merchants where they bought the Chinese kale from and often many of them had no answer. For those who provided this information, it appeared that most of the Chinese kale samples tested were bought from four different whole sale markets in Bangkok and Nakhon Pathom Province located near Bangkok. This information was insufficient to identify the farms where these Chinese kale samples were cultivated. It is anticipated that future studies will be conducted to trace the farms where the vegetables are grown and to identify the factors or farmers' behaviors that contribute to the differences in pesticide detection rate and the MRL exceedance. More importantly, appropriate education such as GAP regarding the proper use of pesticides should be given to these farmers.

This study found an overall incidence of pesticide detection in the Chinese kale studied was 85%. This figure is much higher than the acceptable detection rate in Western or developed countries, such as the USA and European Commission (EC) countries like France, the U.K., Norway and Germany. For instance, the US FDA conducted a monitoring program of vegetables including thousands of domestic samples and imported samples (Granby et al., 2008). Pesticide residues were detected in 30% of the domestic and 21% of the imported vegetables. In Taiwan, between 1997 and 2003, pesticide residues were detected in 14% of 9955 vegetable samples tested (Chang et al., 2005). A survey

study in India found residues of many organophosphorus pesticides (e.g., chlorpyrifos, dimethoate, monocrotophos and profenofos) in 54% of the vegetable samples (Swarnam and Velmurugan, 2013). Recent study from Thailand (Sapbamrer and Hongsibsong, 2014) reported an overall pesticide detection rate in many vegetables purchased from the markets, of 25% (N = 106). This rate is still much lower than the rate of pesticide detection in the Chinese kale that was observed in this study. The difference may be attributed to different types of vegetables, seasons of vegetable cultivation, types of pesticides used and analytical methods employed.

Interestingly, the incidence of pesticide detection of exceeding the MRL in the Chinese kale samples tested observed in this study was 29%, which is considerably high as compared with the incidence reported in developed countries. For examples, the US FDA stated that violations (with pesticide concentration > MRL) were detected in 2% of the domestic and 7% of the imported vegetable samples (Granby et al., 2008). The EU Monitoring Program for pesticides reported that 5% of vegetable samples surveyed had the pesticide residue concentrations exceeded the MRL (Granby et al., 2008). In Asia, a study conducted in Taiwan reported that of 9955 samples, 1.2% was violating the MRL (Chang et al., 2005). Thus, the incidence of pesticide detection of >MRL in our Chinese kale samples at a rate of 29% is exceptionally high when compared with acceptable rate reported in developed countries. However, this incidence rate resembles that found in an Asian country, Pakistan, in which 206 different vegetables were analyzed for 24 pesticides, and 46% had levels exceed the MRL (Parveen et al., 2005). A study from India (Swarnam and Velmurugan, 2013) found that 15% of vegetable samples tested contained pesticide residues that exceeded the MRL values. In addition, the incidence of pesticide detection of >MRL was reported to be 24% in various market vegetables tested in northern region of Thailand (Sapbamrer and Hongsibsong, 2014). This rate of pesticide detection is quite similar to that is reported here. Recently, the Food and Drug Administration (FDA), Ministry of Public Health of Thailand released a report on the monitoring program of pesticides in vegetables and fruits included more than 60,000 samples tested each year (Srithongkum, 2014). The report showed that violations (with pesticide concentration > MRL) detected in vegetables and fruits sold in Thailand were in a range of 4% in year 2013 to 5% in year 2011. These figures reported by the FDA of Thailand were approximately 6 times lower than the incidence rate (29%) of pesticide detection of >MRL found in this study. The contradictory findings are likely to be explained by the difference in methods used in the two survey studies. The survey by the FDA of Thailand was performed by using cholinesterase inhibition assay kit called GT-Test kit. This assay kit is capable of detecting two groups of pesticides namely carbamates (carbofuran and methomyl) and organophosphates (dicrotophos and EPN). However, unlike our current GC–MS/MS assay, the kit GT-Test kit cannot provide a quantitative analysis as most analytical methods, such as UV spectrophotometric assays, LC–MS/MS and GC–MS/MS. As the kit assay is limited in detection to only four individual pesticides, has less sensitivity and does not give a quantitative determination of pesticide concentration, it can underestimate the incidence of violations.

One of possible reasons accounted for such a high incidence rate (29%) of pesticide detection of >MRL in our Chinese kale samples investigated may be related to the short pre-harvest intervals. The pre-harvest interval is described as the time period (in days) between the last pesticide application and a safe harvest of the treated crop (Stephenson et al., 2006). The pre-harvest intervals of pesticides in the Chinese kale samples were unknown in this study. Knowledge of pre-harvest interval is very important for farmers to understand and appreciate. This is because it will give farmers as a guide to when to harvest the crops after the pesticides have been applied. Without this awareness, the farmers may harvest the Chinese kale too soon before the end of pre-harvest interval. This will obviously

result in higher pesticide residues retained in the vegetables than when the vegetables are harvested on or after the recommended pre-harvest interval.

Our findings of a high detection rate (85%) of pesticide residues in the Chinese kale cultivated in Thailand reflect a considerable contamination of pesticides in a commonly consumed vegetable. It is likely to be a consequence of widespread and over-use of pesticide application in Thai agriculture. The high rate of pesticide detection (85%) with high incidence of pesticide detection of >MRL found in this particular vegetable can jeopardize human health as the Chinese kale is popularly eaten by Asian people. In some cuisines, it is eaten as fresh vegetable without cooking. Such high rate of pesticide detection also suggests a great risk of pesticides' toxicity when eating this vegetable. This is simply related to the chance of getting the Chinese kale contaminated with pesticides from the markets is as high as 85%. It could be estimated that almost 9 in 10 Chinese kale bunches bought from local markets in Nakhon Pathom Province, Thailand, will contain detectable pesticides. Exposure to pesticides has been related to increased cancer risks and reproductive dysfunctions in humans (Clostre et al., 2014; Harnpicharnchai et al., 2013; Rönn et al., 2011; Salerno et al., 2014; Uysal et al., 2013). Cooking with heat such as boiling and frying, is likely to destroy or reduce pesticides that contaminated with vegetables and fruits, thus reduce the intake of pesticides into human body (Granby et al., 2008). Blanching and frying of pepper and eggplant fruits resulted in great reduction to almost completely removed (approximately 100%) of the deposited profenofos (Radwan et al., 2005). However, in some dishes, vegetables like Chinese kale are eaten freshly, so the intake of pesticide residues will be high if the vegetables are contaminated and are not washed properly before use. The high incidence (29%) of pesticide detection exceeding the MRL in the Chinese kale samples observed in this study may indicate possible health risks to the Thai population. It was mentioned previously that MRLs are product limits, not safety limits. Thus, if an MRL is exceeded, one can no longer ensure the safety of the product (e.g., vegetable and fruit). However, whether there is a true safety issue or not should be further clarified by carrying out a dietary risk assessment. We have not, however, used the existing results to attempt a dietary risk assessment associated with eating Chinese kale. This is because we plan to perform this task in a follow-up study with a larger sample size.

The findings of detected pesticides in Chinese kale sold in various local markets of Nakhon Pathom Province, Thailand, showed that residue levels of 7 out of 28 pesticides studied were above their respective MRL values established by Thai government authority. These included carbamates (carbofuran), chlorothalonil, metalaxyl, organophosphorus pesticides (chlorpyrifos, dimethoate, profenofos), and pyrethroids (cypermethrin). Of these pesticides detected, metalaxyl (74%), diazinon (42%), dimethoate (41%) and profenofos (25%) are among the most often found in the Chinese kale samples tested. This suggests that these four pesticides are the most popularly used pesticides in the cultivation of Chinese kale in this region of Thailand. The pattern of pesticide use in Thailand may be different in other parts of the country. For example, a study in a northern province of Thailand (Sapbamrer and Hongsibsong, 2014) reported that the most common organophosphorus pesticides detected in vegetable samples from markets were chlorpyrifos (34%), diazinon (19%), parathion-methyl (3%), profenofos (3%), pirimiphos-methyl (3%) and fenitrothion (2%). Chlorpyrifos was the most detected pesticide in the study conducted in a northern Thailand province (Sapbamrer and Hongsibsong, 2014) whereas metalaxyl was the most often detected pesticide in the current study. However, these two studies shared a common finding in which diazinon and profenofos were common pesticides found in vegetables. The present investigation, however, has some limitations as only 28 pesticides were studied. This may not cover all pesticides used in Thailand for cultivation of Chinese kale. Some pesticides such as dithiocarbamates (zineb,

thiram, propineb, maneb and mancozeb) and abamectin, although were reported to be found in Chinese kale, but the analytical methods for these pesticides are not available in our laboratories. Thus, they were not included in the study.

4.3. Washing removes some pesticide residues

Regarding the practical methods to reduce contamination of pesticides in vegetables, washing the vegetables is probably the most simple and economic way to achieve this. Washing has been demonstrated to reduce pesticides that are loosely attached to the surface whereas peeling removes even those that have penetrated the cuticles of the fruits or vegetables (Abou-Arab, 1999; Keikotlhaile et al., 2010). The present study has demonstrated that for washing leaf vegetables such as Chinese kale, washing with running water appeared to be the most effective method to remove pesticides from the vegetable. This running water method involves washing the vegetables under running tap water for 2 min. While washing under water stream, it is also advisable to rub vegetable with your hands as this action helps to remove the contaminated pesticides present on the vegetable surface. In this study, running water method was proven to be the most effective one to reduce the amount of one pesticide profenofos. The results have shown that after washing with this running water method, the concentration of profenofos in the Chinese kale was reduced to 45%. This means at least approximately 55% of pesticides were removed by running water method. The remaining pesticides found after washing were those inside the vegetables and cannot be washed. The running water method was also proven to be superior over other washing methods suggested. These included washing by using NaHCO_3 , vinegar and KMnO_4 . However, this may not always be true as with another pesticide cypermethrin tested, diluted vinegar solution seemed to be more effective in removing cypermethrin than running water, NaHCO_3 and KMnO_4 methods. Differences between results obtained with profenofos and cypermethrin may be related to differences in their chemical characteristics such as pK_a , water solubility and solubility in acids and bases. The effectiveness of washing methods could be dependent upon the type of pesticides. As most pesticides are either weak acids or weak bases, and they have differences in physicochemical properties such as water solubility and tissue binding. When the two pesticides investigated were compared, our results revealed that washing the Chinese kale samples with running water method was more effective to remove profenofos than cypermethrin. This is likely to be explained by the fact that profenofos has higher water solubility (28 mg/L) than that of cypermethrin (0.004 mg/L) (Central Agricultural Pesticides Laboratory, 2015). Conversely, cypermethrin was more effectively removed by washing with vinegar solution than profenofos, which could be due to some other factors such as their relative solubility in acetic acid (vinegar) solution and the relative stability of these compounds in acidic condition. In general, choosing the most effective washing method to remove pesticides from vegetables before consuming is extremely difficult. However, when considered from the economic point of view, the running water method may be the most practical and easy to do. Water is likely to be good and most safe solvent to wash out every pesticide by dissolving and washing them away. Our washing results were consistent with previous reports in which they have demonstrated that washing vegetables with water can remove the pesticide residues but not totally all of them. For examples, by washing the tomato fruits, chlorpyrifos residues were reduced by 41 to 44% (Rani et al., 2013). Washing followed by boiling reduced the chlorpyrifos residues from 89 to 91% whereas peeling removed residues from 62 to 64% (Rani et al., 2013). Chavarri et al. (2005) found that washing reduced chlorpyrifos residues in asparagus by 24%, and 92% after cooking. Washing with water as well as the cooking process helped to eliminate most of the pesticide residues from the potato tubers (Soliman, 2001). By washing cucumbers under running tap water for 1 min, it was shown to remove carbaryl residues by 33% (Hassanzadeh et al., 2010).

5. Conclusion

It is evident that for most Asian developing countries, an effort has been made to regulate pesticides used in agriculture. Even though, routine monitoring programs of pesticide residues are not fully functioned, there is awareness of pesticide contamination in foods and their toxicity among government authorities. This article is the first one of our series and we hope that our study will further stimulate this awareness, especially in Thailand and Asian neighboring countries. The research should also provide useful information for establishing the routine monitoring programs. Furthermore, consumers should be informed about the levels of pesticide contamination in vegetables and fruits. The farmers as producers must also be aware of pesticide contamination levels and the potential dangers of pesticide contamination in food commodities. Laws on forbidden pesticides should be more strictly enforced. In parallel, education and awareness building among farmers about safe pesticide use and pre-harvest interval for crops, is essential. These measures will ensure greater public safety in the daily consumption of vegetables and fruits.

The present study provides scientific evidence of detected significant residues of many pesticides in the commonly consumed vegetable Chinese kale sold in Thailand. The important findings are the observed high rate of pesticides detected and considerable high incidence of pesticide detection of greater than their MRL in this commonly consumed vegetable. It draws attention for Thailand agriculture authorities to improve the management of pesticide use and control as well as the education and promotion of safe pesticide use. The findings suggest that monitoring of pesticide residues in vegetables and fruits should be performed on a routine basis. The results also assist in risk assessment of consumers' exposure to the harmful pesticide residues. Proper washing of vegetables before using or consuming is equally important, thus public education on washing is very crucial. Consumers, cooks and restaurant owners must be fortified to wash vegetables with a lot of running water before use as this will considerably reduce the intake of pesticides. Use of so called biopesticides is an alternative approach to avoid using synthetic harmful pesticides. These biopesticides are not synthetically produced, but are derived from natural materials such as botanical pesticides, and microbial/biological agents. Thus, the biopesticides are considered to be safer than commonly used pesticides (Chandler et al., 2008; Gakuya et al., 2013; Villaverde et al., 2014).

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Conflict of interest

The authors declare that they have no conflict of interest.

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